Fidma Mohamed Abdelillah, Data scientist

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Education

2023 – 2026 **Ph.D., Gustave Eiffel University** Marne-La-Vallée, France.
Thesis title: *Deep learning applied to the Stuctural health monitoring of civil structures.*
Research focuses on integrating deep learning into SHM methodologies
Topics include damage identification, structural assessment, predictive modeling, and optimization of sensor-related strategies

2021 – 2023 **M.Sc. in Computer Science, ESI SBA (École Supérieure en Informatique),** Sidi Bel Abbès, Algeria.
Master's Thesis (internship-based):Studying and Comparing the existing approaches for predictive maintenance in industry 4.0

2018 – 2023 **Computer Science, ESI SBA Engineering School** SBA, Algeria.
Specialization : *information systems engineering.*

Employment History

Apr 2023 – Nov 2023 **Research AI intern** CESI Lineact, LYON, France.
Thesis title: *Hybrid predictive maintenace system based on ontologies and machine learning in industry 4.0*
Building and optimizing ML models for Predictive maintenance system,
Integrating ontologies into AI systems for semantic understanding.

Feb 2021 – Apr 2023 **Mobile developer,** Fennec Dev.
Cross-platform Development: Building apps for multiple platforms with React Native or Flutter.

Sep 2021 – Nov 2021 **Information System Network Intern** Centre des Technique spatiale, Arzew, Algeria.
Assisting with network management and configurations.

Research Publications

Journal Articles

1. M. A. F. P. Student, F. S. Dr, and J.-F. B. Prof., "Deep learning techniques for damage detection in structural health monitoring: A review," *Structural Engineering International*, vol. 0, no. 0, pp. 1–17, 2025. DOI: 10.1080/10168664.2025.2584752. eprint: https://doi.org/10.1080/10168664.2025.2584752.

Conference Proceedings

1. M. A. Fidma, J.-F. Bercher, and F. Schmidt, "Intelligent Aggregation of Single-Sensor Classifiers for Enhanced Structural Health Monitoring Networks," in *8th International Conference on Machine Learning for Networking (MLN'2025)*, Selma Boumerdassi, Nour El Houda Yellas, Éric Renault, Paris, France, Dec. 2025. URL: https://hal.science/hal-05451332.
2. M. A. Fidma, J.-F. Bercher, and F. Schmidt, "Low-Complexity Approach to Intelligent SHM by Combining Machine Learning Models Using Single-Sensor Data," in *Proceedings of the 15th International Workshop on Structural Health Monitoring*, I. Destech Publications, Ed., Chang, Fu-Kuo and Guemes, Alfredo, Stanford University, Stanford, CA, United States, Sep. 2025, pp. 983–990. DOI: 10.12783/shm2025/37384.

3 M. A. Fidma, F. Schmidt, and J.-F. Bercher, "Damage detection of civil structures through machine learning approaches," in *International Association for Bridge and Structural Engineering Congress Ghent 2025*, The International Association for Bridge and Structural Engineering (IABSE), Gand, Belgium, Aug. 2025. URL: https://hal.science/hal-05195040.

4 F. M. Abdelillah, H. Nora, O. Samir, and S. M. Benslimane, "Predictive Maintenance Approaches in Industry 4.0: A Systematic Literature Review," in *2023 IEEE International Conference on Enabling Technologies: Infrastructure for Collaborative Enterprises (WETICE)*, Paris, France: IEEE, Dec. 2023, pp. 1–6. DOI: 10.1109/WETICE57085.2023.10477802.

5 F. M. Abdelillah, H. Nora, O. Samir, and B. Sidi-Mohammed, "Hybrid Data-Driven and Knowledge-Based Predictive Maintenance Framework in the Context of Industry 4.0," in *Proceedings*, ser. Lecture Notes in Computer Science, vol. 14396, Sousse, Tunisia: Springer Nature Switzerland, Nov. 2023, pp. 319–337. DOI: 10.1007/978-3-031-49333-1_23.

Service and Teaching

Teaching assistant

2024 **Apprentissage de la programmation avec Python et C** , ESIEE PARIS.

2025 \2026 **Python 1 : programmation avancée**
Python 2 : manipulation de données
Introduction à la datascience , ESIEE PARIS.

2026 **Time Series & Machine Learning**, ESIEE PARIS.

workshop mentor

2024 Created and mentored a workshop titled *PyCivil*, focused on Python fundamentals for civil engineers, including data manipulation, visualization, and basic scientific computing, EMGCI, UGE.

Skills

Data Analysis & Machine Learning: Strong theoretical knowledge and hands-on experience with Python libraries. Limited experience with R.

Deep Learning: Solid understanding of neural networks architectures. Experienced with PyTorch, TensorFlow, and Keras.

Digital Signal Processing: Signal filtering, feature extraction, FFT, and sensor data analysis.

DBMS: MySQL, PostgreSQL, MongoDB (NoSQL).

MLOps: Experience with Docker, MLflow, GitHub Actions for CI/CD, model versioning, and deployment pipelines. Familiar with cloud platforms (AWS, GCP) for scalable ML deployment.

IoT: Arduino and Raspberry Pi projects, deploying lightweight AI models to edge devices. Data collection and preprocessing for SHM applications.

Version Control Systems (VCS): Git and GitHub.

languages

- Frensh, C1
- English, C1
- Arabic, Native

Interests

- Chess
- Athletics
- Human History